Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/28/2015
PI Name:	Pluth, Janice Ph.D.		
Project Title:	Impact of Age, Genetics Variants and High LET Track Structure on Mammary Cancer Risk Estimates		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/18/2013	**End Date:**	01/17/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 1/17/2017 per PI (Ed., 2/23/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ13HA96I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A key NASA objective is to obtain quantitative data defining the risk of carcinogenesis following galactic cosmic ray (GCR) exposure. Current radiation quality factors used to determine cancer risk for space radiation do not provide the accuracy needed, and debate exists as to the relevance of the experimental studies they're based upon. To formulate better risk estimates, key data is needed including how age, genetic susceptibility, and radiation quality affect carcinogenic potential in irradiation sensitive tissues. Towards this aim, we will use a novel resource of a large collection of normal mammary epithelial cell strains, from women aged 16 to 80, including many in the age of astronauts, to address these important questions. We will obtain quantitative data regarding radiation quality effects; ROS (reactive oxygen species) and DNA damage signaling, radiation-induced centrosome defects, and enrichment of stem cell populations using this cohort of strains from various aged individuals. In addition, we will also investigate		

	genetic susceptibility, including how loss of p16 expression, a sporadic event in individual cells of normal women in vivo and often observed in breast tumors, affects radiation quality effects and centrosome aberration frequency. Additionally we will analyze strains containing a BRCA1 or an ATM heterozygote genetic background, representing ~0.32 and ~1% of the normal population respectively, to determine how these genetic variants may make them more sensitive to high LET (linear energy transfer) exposures. These studies will provide critical data, which can be used to model how genetic susceptibility, age, and radiation quality interplay in defining the cancer risk of mammary cells to high LET radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Brief summary of past year's research accomplishments (November 2015): In 2015 we undertook an additional radiation run at Brookhaven labs (spring, N15A) to obtain additional samples lost due to technical and/or shipping issues. During the run we exposed cells to Si 93 MeV/u, Ti300 MeV/u, and Fe 1000 MeV/u ions. Additional cesium experiments were also performed in this year to complete the low LET portion of the study. Due to the extremely labor intensive tissue culture involved in this study, we are still in the process of thawing some of the exposed samples from N15A and growing them out to score for centrosome defects and stem cell population changes. Given most other studies to date have concentrated on defining genomic instability at early times post exposure, we have chosen to define centrosome aberrations at a later time, weeks post exposure. This will provide additional data regarding how radiation exposure persistently affects cells, using a timepoint that may be more relevant and useful in modeling cancer risk. Thus we have thawed samples frozen down at the run and continued to culture them for an additional 9 -12 days to determine how centrosome levels are affected by the age of the individual and the radiation quality of the exposure. We have also defined the percentage of stem/progenitor cells in these various exposed populations to define if radiation changes influences the growth of these key cells and how age may impact this. Lastly, to gain further information from this cohort of samples, we have also performed assays using a smaller subset to look at gene expression and miRNA expression following high or low LET radiation exposure. Overall summary: In total our studies thus far have yielded robust results that would be invaluable in modeling long-term effects of high LET exposure. As age and individual susceptibility are critical factors that impact cancer risk modeling, we have used power calculations to determine the lowest number of samples that would need to be assayed to balance achieving significance, while minimizing costs. We have completed work for 70% of the strains and based on this have extrapolated that we will need an additional year of funding to complete this work. To this end we have requested a no cost extension to allow us to do so. The results of these studies will be crucial to quantitatively factor in the influence of age and radiation quality on genomic instability and stem cell populations, endpoints that have been linked to cancer. Thus these studies will provide data useful for modeling the risk of mammary cancer from space radiation exposures and decrease the uncertainties due to age and genetic variation.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Sridharan DM, Asaithamby A, Bailey SM, Costes SV, Doetsch PW, Dynan WS, Kronenberg A, Rithidech KN, Saha J, Snijders AM, Werner E, Wiese C, Cucinotta FA, Pluth JM. "Understanding cancer development processes after HZE-particle exposure: roles of ROS, DNA damage repair and inflammation." Radiat Res. 2015 Jan;183(1):1-26. Review. http://dx.doi.org/10.1667/RR13804.1 ; PubMed PMID: 25564719 , Jan-2015